

Methodologies for the correction of Garbage codes in mortality records and their applicability in microsimulation models: Protocol for a scoping review

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2026-07-31

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1 Abstract

Objective: To systematically map and synthesize the scientific evidence on the quantitative methodologies used for the identification, cleaning and redistribution of junk codes in population-based mortality registries in low- and middle-income countries (LMICs), evaluating their technical applicability for the parameterization of microsimulation models in health.

Inclusion criteria: This review will consider studies utilizing population-based mortality records or vital statistics from low- and middle-income countries (Population). Eligible literature must describe or apply traditional statistical methods, multiple imputation, Bayesian models, record linkage, or machine learning algorithms to correct ill-defined causes of death and “garbage codes” (Concept). The context includes health information systems characterized by scarce or heterogeneous data, specifically where the feasibility of disaggregating results to the individual level to parameterize stochastic microsimulation models is analyzed (Context).

Methods: This scoping review will follow the JBI methodology and the PRISMA Extension for Scoping Reviews (PRISMA-ScR). Comprehensive searches will be performed across PubMed, Scopus, Web of Science, LILACS, and SciELO, supplemented by grey literature from the WHO, PAHO, and IHME. To ensure reproducibility, the workflow will be executed within the R statistical environment. We will utilize `litsearchr` for search syntax optimization, `synthesizr` for deduplication, and `revtools` for blinded screening supported by latent Dirichlet allocation topic clustering. Discrepancies will be resolved by discussion or a third reviewer. Following paired data extraction, findings will be synthesized narratively and visualized using `ggplot2` matrix heatmaps and `igraph` flow networks.

Keywords: health information systems; causes of death; vital statistics; data quality; machine learning algorithms; proportional redistribution; Computer simulation.

2 1. INTRODUCTION

Mortality statistics derived from Civil Registration and Vital Statistics systems serve as the bedrock of global epidemiological surveillance and public health planning. However, their utility is frequently undermined by medical certification errors, primarily the assignment of deaths to “garbage codes”, defined as International Classification of Diseases codes that represent ambiguous, non-specific, or biologically implausible underlying causes¹. Because raw mortality data frequently lack diagnostic precision, the development and application of indirect statistical methodologies to correct and redistribute cause of death statistics remain a crucial public health priority^{2,3}. This is particularly alarming given the escalating toll of non-communicable diseases in low- and middle-income countries, whose rapid dynamics require highly reliable longitudinal data⁴.

To mitigate these systematic biases, a diverse array of correction and redistribution methodologies has evolved over the past two decades^{5,6,7}. At their core, modern redistribution strategies are typically classified into four distinct mathematical approaches: multiple cause analysis, negative correlation, impairment reallocation, and proportional redistribution⁷. Beyond these foundational methods, researchers have developed advanced computational techniques to improve data utility, such as subnational regression-based models validated through simulations⁸ and reproducible four-step probabilistic algorithms⁹. More recently, cutting edge spatial and machine learning approaches have emerged to handle localized data. These include the use of spatial scan statistics¹⁰ and Local Moran’s I autocorrelation to adjust for spurious geographical clusters of unexplained mortality¹¹. Furthermore, researchers are increasingly utilizing non-linear random forest algorithms¹², decision trees¹³, and the application of non-parametric coarsened exact matching using multiple causes of death to empirically redistribute garbage codes without relying on arbitrary functional assumptions¹⁴.

Despite this remarkable proliferation of analytical techniques, current literature remains fragmented, lacking a systematic mapping to consolidate these diverse mathematical assumptions and evaluate their techni-

cal transferability to individual level microsimulation models¹⁵. Unlike aggregated demographic models that rely on macroscopic trends, microsimulations simulate single life trajectories and require parameterization with highly disaggregated baseline mortality risks¹⁶. Currently, no evidence synthesis exists on how to adapt or transfer redistribution methodologies to generate the granular datasets suitable for stochastic microsimulation¹⁷. Consequently, this scoping review will map the current state of the art⁹, evaluate the technical feasibility of these approaches within simulation environments¹⁶, and provide a framework for future public health mathematical modeling¹⁸.

2.1 1.1 Research Questions

1. What analytical methodologies (statistical, probabilistic, deterministic, linkage-based or machine learning) have been described in the literature for the correction and redistribution of “garbage codes” in mortality records?
2. How are they classified and what types of “garbage codes” (poorly defined causes of ICD chapters, intermediate or terminal causes) are the main objective of each identified methodology?
3. What is the documented level of applicability or compatibility of these correction methods to structure databases aimed at the development, calibration and validation of microsimulation models in health?
4. What are the main methodological challenges, limitations and advantages reported when transferring or adapting analytical correction algorithms from the population (macro) level to the individual level (micro)?

3 2. ELIGIBILITY CRITERIA

3.1 2.1 Population

This component defines eligible data records, populations, and geographic or demographic coverages.

- **Inclusion Criteria:**

- **Data Type:** Studies utilizing population-based mortality databases, vital statistics systems, official national or subnational health statistics registers, or civil registration and vital statistics (CRVS) systems.
- **Geography and Income Level:** Research focusing on populations within low- and middle-income countries (LMICs) as defined by historical or current World Bank classifications (e.g., regions within Latin America and the Caribbean, Sub-Saharan Africa, and South Asia).
- **Technical Exception:** Global or multinational studies (e.g., Global Burden of Disease [GBD] analyses), provided they independently disaggregate methodologies and data applicable to LMIC geographies.
- **Demography:** Records tracking general human mortality without restrictions on age groups (including neonatal, infant, adult, or geriatric mortality) or sex and gender.

- **Exclusion Criteria:**

- Clinical studies restricted to hyper-specific hospital specimens or controlled trials that do not utilize—or aim to correct—population-based records or broader epidemiological information systems.

3.2 2.2 Concept

This component delineates the core methodologies, algorithms, and analytical tools evaluated in this review.

- **Inclusion Criteria:**

- Studies describing, evaluating, validating, or implementing quantitative methods to identify, clean, reclassify, or redistribute “garbage codes” (levels 1 to 4) or ill-defined causes of death (as classified under ICD-10/ICD-11 Chapter XVIII, or ICD-9 equivalents).
- Methodologies within the following analytical categories are eligible:
 - * **Traditional statistics and mathematical demography:** Proportional redistribution (simple or stratified by age and sex), fractional algorithms, and indirect demographic adjustment methods⁵.
 - * **Advanced statistical modeling:** Multiple imputation of missing or non-specific data, spatial or spatiotemporal regression modeling, and empirical or hierarchical Bayesian approaches (e.g., CODEm or CoDcorrect analytical algorithms)^{7,8,19}.
 - * **Data integration (record linkage):** Deterministic or probabilistic algorithms linking mortality records with complementary databases (e.g., hospital discharge records, automated verbal autopsies, or epidemiological surveillance systems)^{20,21}.
 - * **Artificial intelligence and machine learning:** Data-driven predictive architectures (e.g., decision trees, gradient boosting, random forests, or other data-mining frameworks) applied to classify or predict underlying causes of death masked by non-specific codes.
- **Exclusion Criteria:**
 - Studies analyzing mortality data quality in a purely descriptive manner (e.g., reporting isolated completeness percentages or garbage code prevalences) without implementing or evaluating an active mathematical or algorithmic correction method.
 - Research focused solely on correcting numerical under-registration (omission of death capture), unless the methodology explicitly integrates cause-of-death redistribution.

3.3 2.3 Context

This component defines the data environments and final fields of application toward which the evidence synthesis is oriented.

- **Inclusion Criteria:**
 - **Data-Scarce or Heterogeneous Settings:** Research situated in contexts with weak or fragmented health information systems characterized by low medical certification coverage, high regional heterogeneity, or small-area spatiotemporal misalignments.
 - **Microsimulation Applicability:** Studies evaluating or discussing the impact of correction techniques on individual-level microdata disaggregation, the generation of consistent mortality risk matrices, individual trajectory estimations, or parameterizing inputs for health microsimulation platforms.
 - **Emergent Methodological Scope:** Studies originally operating at a macro or aggregate level, provided they detail the mathematical assumptions or individual-record classification algorithms necessary to calibrate and parameterize stochastic microsimulations or agent-based models.
- **Exclusion Criteria:**
 - Methodologies operating exclusively at an irreducible macro-population level whose designs preclude individual-level micro-disaggregation, thereby lacking utility for agent-based or individual-level simulations.

3.4 2.4 Sources of Evidence

- **Publication Types:** Peer-reviewed original research articles published in indexed journals, methodological book chapters, and authoritative institutional gray literature (e.g., technical reports from the WHO, PAHO, or IHME).

- **Timeframe:** Material published from 1996 onwards, capturing the global introduction and widespread adoption of the Tenth Revision of the International Classification of Diseases (ICD-10), which consolidated the term “garbage codes.”
- **Languages:** Studies published in English, Spanish, and Portuguese, reflecting the substantial volume of methodological literature generated in Latin America (particularly Brazil).

4 3. METHODS

4.1 3.1 Search Strategy

The search strategy will locate published academic literature and online grey literature. The search will focus on evidence published from 1996 onwards, corresponding to the global implementation and widespread adoption of the International Classification of Diseases, Tenth Revision (ICD-10)—a timeframe during which the term “garbage codes” became conceptually consolidated.

- **Academic Literature:** To identify academic literature, a three-step search strategy will be developed in collaboration with a medical librarian:
 1. **Initial Limited Search:** An initial search of MEDLINE (Ovid) will be undertaken to analyze text words in titles and abstracts, Medical Subject Headings (MeSH), and author-supplied keywords from relevant articles.
 2. **Full Search Expansion:** A comprehensive search strategy will be developed using these identified terms, then translated and adapted for Scopus, Web of Science, LILACS, and SciELO. The complete search syntax for MEDLINE (Ovid) is presented in Appendix I.
 3. **Reference Hand-Searching:** The reference lists of all selected articles and relevant systematic or scoping reviews will be hand-searched to identify primary sources missed by the electronic search.
- **Grey Literature:** To locate technical reports and eligible grey literature, a targeted list of national and international health organization websites will be compiled following the two-step Google search methodology.
 1. **Preliminary Term Identification:** A preliminary Google search will be conducted to identify key text words, accounting for highly variable terminology such as “unspecific causes” or “ambiguous causes.”
 2. **Structured Execution:** Ten unique structured Google searches will be executed using distinct keyword combinations representing the population, concept, and context (PCC) of interest.

The first 100 results of each structured search will be screened. Identified organizational websites will be manually evaluated for technical reports. Finally, vital registration epidemiologists, public health officials, and key collaborators with expertise in health information systems within LMICs will be consulted to identify any additional missing sources.

4.2 3.2 Data Management and Search Automation

To ensure the methodological reproducibility and transparency required by the PRISMA-ScR guidelines, the information retrieval process will be systematized within the R statistical environment (v4.5 or higher). The `litsearchr` package will be employed for the algorithmic optimization of Boolean syntax via term co-occurrence networks. Indexed databases will be interrogated programmatically through Application APIs using the `rentrez` and `rscopus` libraries. Data consolidation and duplicate removal will be managed via the `synthesizr` package using fuzzy string-matching models, supplemented by manual verification of borderline cases. Finally, title and abstract screening will be conducted through the interactive graphical interface of the `revtools` package, supported by Latent Dirichlet Allocation (LDA) algorithms for topic modeling to optimize evidence selection in accordance with the PCC framework.

4.3 3.3 Data Extraction and Charting

To ensure integrity, traceability, and reproducibility, we will conduct data extraction programmatically within the R statistical environment (v4.5 or higher). This automated approach mitigates the transcription errors inherent in manual processes and ensures a standardized workflow. We have designed an electronic data charting form to capture information across six dimensions:

- **Study metadata:** Author, year of publication, country, and study design.
- **Geographic and health system context:** Challenges encountered during implementation, such as inadequate medical training or certification capacity, and recommendations for systemic improvement²².
- **Mortality database characteristics:** Specific baseline registry quality metrics used to evaluate the initial validity of the input data. A prime example is the five-star rating system, which objectively scores mortality data based on demographic completeness and the prevalence of major garbage codes⁷, as well as the algorithmic assessment of missing or unexpected values in core demographic fields, which serves as a critical proxy for systemic data quality issues in highly deprived municipalities¹².
- **“Garbage code” typology:** The specific definition of a garbage code applied by the authors³, alongside the age- and sex-specific epidemiological profiles of these codes, to understand the clinical nature of the missing data at different life stages⁷.
- **Correction methodologies:** The type of redistribution model used, the algorithmic and software ecosystems utilized, and the specific performance metrics reported to demonstrate data quality improvement¹⁷.
- **Microsimulation applicability:** The explicit capacity of the methodology to generate highly disaggregated microdata and handle parameter uncertainty for stochastic predictive modeling¹⁶.

This form will be piloted with a random sample of three included studies to refine consistency and ensure variable granularity. Two reviewers will perform the extraction independently, resolving discrepancies through consensus. The resulting data will be consolidated into a relational structure (e.g., tibble or data.frame) to facilitate transparent downstream analysis. The full extraction schema (exported as JSON/CSV) and corresponding R scripts will be made available via the Open Science Framework (OSF) upon study completion. For a detailed breakdown of the variables and the classification framework utilized, refer to the Data Charting Instrument (Appendix II).

4.4 3.4 Data Analysis, Synthesis, and Presentation

This scoping review adheres to the Joanna Briggs Institute methodology. We will conduct all data processing and visualization programmatically within the R statistical environment.

- **Descriptive Quantitative Analysis:** We will calculate absolute frequencies and percentages to characterize the literature by chronological distribution, geographic reach, taxonomic frameworks, cause categories, and analytical techniques. These analytical techniques will be categorized by their computational complexity, ranging from foundational deterministic methods to advanced predictive modeling, machine learning, and spatial analysis.
- **Thematic Narrative Synthesis:** A narrative will accompany the extracted data, addressing the magnitude of garbage codes in data-scarce settings, the methodological diversity of the algorithms, and the conceptual transferability of these methods to preserve microdata validity for agent-based stochastic models. This includes addressing how uncorrected garbage codes disproportionately mask the true mortality burden in populations with higher social deprivation²³ and the risk of inappropriate centralization when macro-level regression algorithms are applied to highly localized municipal datasets²⁴.

- **Mapping and Visualization:** To visualize the structure and knowledge gaps of the field, we will generate advanced graphics using ggplot2²⁵ and igraph²⁶. This will include structured evidence tables cross-referencing correction methods with their input data requirements, two-dimensional heatmaps mapping analytical methods against specific ICD Chapters, and flow network diagrams illustrating the technical pathway required to connect traditional vital statistics with microsimulation parameterization.

5 Conflicts of Interest

The authors declare no competing interests. This scoping review was conducted for academic and methodological purposes only, with no commercial, financial, or institutional support that could be construed as a potential conflict of interest.

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7 Appendices

7.1 Appendix I: Search strategy

Database: MEDLINE (via Ovid)

Search conducted on: July 2026

Planned Limits: * **Date restrictions:** from 1996 to present to capture historical development of algorithms like GBD. * **Language restrictions:** None at search level (any required filters based on reviewer capacities will be applied during the programmatic screening stage). * **Document types:** All source types (including technical notes, methodology papers, and electronic articles).

Line	Search Query (Terms, Truncations, and Syntax)	Conceptual Block / Target
#1	exp Mortality/ or exp Cause of Death/	PCC: Population / Focus (Core mortality registry filters)
#2	(mortality data* or death certificate* or vital statistic* or vital registration or CRVS).tw,kf,ot.	Keywords for vital statistics and national death registry systems
#3	1 or 2	Result: Core Mortality Data
#4	(garbage code* or ill-defined or misclassified or misclassification* or undefined cause* or ill-defined cause* or unspecific cause* or vague code* or redistribution algorithm*).tw,kf,ot.	PCC: Concept (Block A) - Specific typology of Data Errors (“Garbage Codes”)
#5	(data cleaning or data correction or diagnostic accuracy or underlying cause of death).tw,kf,ot.	Data quality attributes specific to mortality databases
#6	4 or 5	Result: Garbage Code Concepts
#7	exp Algorithms/ or exp Models, Statistical/ or exp Computer Simulation/	MeSH Indexing for Analytical Methods

Line	Search Query (Terms, Truncations, and Syntax)	Conceptual Block / Target
#8	(demographic redistribution or multiple imputation or MICE or Bayesian model* or Markov chain* or machine learning or random forest* or neural network* or artificial intelligence or algorithmic correction or fractional assignment).tw,kf,ot.	PCC: Concept (Block B) - Specific statistical and computational correction frameworks
#9	(microsimulation* or micro-simulation* or agent-based model* or stochastic simulation* or individual-level dynamic* or synthetic population*).tw,kf,ot.	PCC: Context / Transferability - Target micro-level application architectures
#10	7 or 8 or 9	Result: Methodological Ecosystem
#11	exp “Developing Countries”/	MeSH Indexing for LMICs
#12	(low income country or low-and-middle income countr* or LMIC or LMICs or developing nation* or resource-constrained setting* or transitional econom*).tw,kf,ot.	Geographic Context Keywords (Based on World Bank specifications)
#13	(Africa* or Asia* or South America* or Central America* or Latin America*).tw,kf,ot.	Broad regional geographic filters
#14	11 or 12 or 13	Result: Developing Context (LMIC)
#15	3 and 6 and 10 and 14	COMBINED STRATEGY (Boolean Intersection)

7.2 Appendix II: Data extraction instrument

Block	Variable	Description / Purpose	Suggested Format / Values
I. Identification & Metadata	Study ID	Unique identifier assigned by R workflow	Numeric (e.g., 001)
	Bibliographic Data	Author, year, title, and journal/institution	Text
	Evidence Type	Original article, technical report, or method book	Dropdown
II. Geographic & Health Context	Geographic Scope	Country, region, or sub-national area	Text
	Economic Classification	World Bank income level	Dropdown (Low, Lower-Middle, Upper-Middle)
	CRVS System Status	Description of Civil Registration and Vital Statistics status	Text

Block	Variable	Description / Purpose	Suggested Format / Values
III. Mortality Database	Data Periodicity	Years covered by mortality records	Text / Numeric
	Data Volume	Sample size (number of deaths analyzed)	Numeric
	Primary Source	Origin of data (e.g., forensic, hospital, surveys)	Text
	ICD Framework	Version of the ICD utilized (ICD-9, ICD-10, ICD-11)	Dropdown
IV. Junk Code Typology	Definition Framework	Criteria used to identify “junk” (e.g., GBD list)	Text
	Evaluated ICD Codes	Specific codes being targeted for correction	Text
	Baseline Magnitude	Percentage/volume of junk codes in original records	Percentage (%)
V. Correction Methodology	Central Technique	Algorithmic classification	Dropdown (e.g., MICE, Bayesian, ML)
	Theoretical Assumptions	Underlying mathematical principles/assumptions	Text
	Software Ecosystem	Tools used (R, SAS, Stata, etc.)	Text
VI. Microsimulation Applicability	Uncertainty/Bias Report	Reported confidence intervals or limitations	Yes / No / Partial
	TRL (Technology Readiness Level)	Maturity of the development method (1-4)	Scale (1-4)
	Output Format	Aggregate vs. individual-level data output	Dropdown
	Demographic Granularity	Consistency at sub-group/small-area levels	Scale (1-5)
	Parametrization Potential	Capability to feed agent-based/microsimulation models	Scale (1-5)
	Reviewer Notes	Professional notes on applicability and transferability	Free text